Homework 1

Vasilii Slyshkin
vslyshkin@connect.ust.hk

Problem 1

Let $A \in \mathbb{R}^{m\times n}$ be given, and define

$$B := \big\{b \in \mathbb{R}^m : \{x : Ax = b,\, x \geq 0\} \text{ is nonempty}\big\}.$$

Equivalently, $B = \{Ax : x \geq 0\}$.

Let $b_1, b_2 \in B$ and $\lambda \in [0,1]$. Then there exist $x_1, x_2 \geq 0$ with $Ax_1 = b_1$ and $Ax_2 = b_2$. Set $x := \lambda x_1 + (1-\lambda)x_2$. Then $x \geq 0$ and

$$Ax = \lambda Ax_1 + (1-\lambda)Ax_2 = \lambda b_1 + (1-\lambda)b_2.$$

Hence $\lambda b_1 + (1-\lambda)b_2 \in B$, so B is convex.

Problem 2

Fix $A \in \mathbb{R}^{m\times n}$ and $c \in \mathbb{R}^n$, and let B be as in Problem 1. For $b \in B$ define

$$z(b) := \min\big\{c^\top x : Ax = b,\, x \geq 0\big\}.$$

Let $b_1, b_2 \in B$ and $\lambda \in [0,1]$, and write $b_\lambda := \lambda b_1 + (1-\lambda)b_2 \in B$. Let x_1, x_2 be minimizers for b_1, b_2 respectively. Then $x_\lambda := \lambda x_1 + (1-\lambda)x_2$ is feasible for b_λ, hence

$$z(b_\lambda) \leq c^\top x_\lambda = \lambda c^\top x_1 + (1-\lambda)c^\top x_2 = \lambda z(b_1) + (1-\lambda)z(b_2).$$

Thus z is convex on B.

Problem 3

Let $A \in \mathbf{S}^n$, $b \in \mathbb{R}^n$, $c \in \mathbb{R}$, and

$$S := \big\{x \in \mathbb{R}^n : x^\top Ax + b^\top x + c \leq 0\big\}.$$

Assume $A \succeq 0$. The quadratic $f(x) := x^\top Ax + b^\top x + c$ has Hessian $\nabla^2 f(x) = 2A \succeq 0$ for all x, so f is convex. Every sublevel set of a convex function is convex, hence $S = \{x : f(x) \leq 0\}$ is convex.

The converse is false: convexity of this particular S does not force $A \succeq 0$. Take $A = -I$, $b = 0$, $c = 0$. Then

$$f(x) = -\|x\|_2^2 \leq 0 \quad \text{for all } x,$$

so $S = \mathbb{R}^n$, which is convex, while A is not positive semidefinite.

Problem 4

(a)

Assume C is a convex cone. If $x, y \in C$, then $\frac{1}{2}x + \frac{1}{2}y \in C$ by convexity, and therefore

$$x + y = 2\Big(\frac{1}{2}x + \frac{1}{2}y\Big) \in C$$

because C is a cone. Thus $C + C \subset C$.

Conversely, assume C is a cone with $C + C \subset C$. Let $x, y \in C$ and $\lambda \in [0, 1]$. Then $\lambda x \in C$ and $(1 - \lambda)y \in C$, so

$$\lambda x + (1 - \lambda)y \in C + C \subset C.$$

Hence C is convex.

(b)

Let C_1, C_2 be convex cones containing the origin.

$C_1 + C_2 = \operatorname{conv}(C_1 \cup C_2)$**.** If $x_i \in C_i$, then $2x_i \in C_i$, and

$$x_1 + x_2 = \frac{1}{2}(2x_1) + \frac{1}{2}(2x_2) \in \operatorname{conv}(C_1 \cup C_2).$$

Thus $C_1 + C_2 \subset \operatorname{conv}(C_1 \cup C_2)$.

For the reverse inclusion: $0 \in C_2$ gives $C_1 = C_1 + \{0\} \subset C_1 + C_2$, and likewise $C_2 \subset C_1 + C_2$, so $C_1 \cup C_2 \subset C_1 + C_2$. The sum of convex sets is convex, hence $\operatorname{conv}(C_1 \cup C_2) \subset C_1 + C_2$.

$C_1 \cap C_2 = \bigcup_{\alpha \in [0,1]} \big(\alpha C_1 \cap (1 - \alpha)C_2\big)$**.**

We first claim that

$$\alpha C_1 \cap (1 - \alpha)C_2 = C_1 \cap C_2 \qquad \text{for all } \alpha \in (0, 1).$$

If $x \in C_1 \cap C_2$, then $x = \alpha(x/\alpha)$ with $x/\alpha \in C_1$, and $x = (1 - \alpha)\big(x/(1 - \alpha)\big)$ with $x/(1 - \alpha) \in C_2$, so $x \in \alpha C_1 \cap (1 - \alpha)C_2$. Conversely, if $x = \alpha x_1 = (1 - \alpha)x_2$ with $x_i \in C_i$, then $x \in C_i$ because each C_i is a cone, so $x \in C_1 \cap C_2$.

If $\alpha \in \{0, 1\}$, then $\alpha C_1 \cap (1 - \alpha)C_2 = \{0\} \subset C_1 \cap C_2$. Combining these cases,

$$\bigcup_{\alpha \in [0,1]} \big(\alpha C_1 \cap (1 - \alpha)C_2\big) = C_1 \cap C_2.$$

Problem 5

1.

We show that $\{x : Ax \le b,\, x \ge 0\}$ is nonempty if and only if $\{y : A^\top y \ge 0,\, y \ge 0,\, b^\top y < 0\}$ is empty.

($\Rightarrow$) Suppose $Ax \le b$, $x \ge 0$, and $A^\top y \ge 0$, $y \ge 0$, $b^\top y < 0$. Then $y^\top Ax \le y^\top b < 0$. But $y^\top Ax = (A^\top y)^\top x \ge 0$, a contradiction.

($\Leftarrow$) Let

$$K := \{Ax + s : x \ge 0,\, s \ge 0\},$$

a finitely generated (hence closed) convex cone. The system $Ax \le b$, $x \ge 0$ is infeasible if and only if $b \notin K$. By the strict separating hyperplane theorem there exist $y \in \mathbb{R}^m$ and $\alpha \in \mathbb{R}$ such that

$$y^\top z \le \alpha \quad \forall z \in K, \qquad y^\top b > \alpha.$$

Since $0 \in K$ we have $\alpha \ge 0$, hence $y^\top b > 0$. Also $y^\top(Ax + s) \le \alpha$ for all $x, s \ge 0$, so $A^\top y \le 0$ and $y \le 0$ (otherwise some ray of K would make $y^\top z$ arbitrarily large). Replacing y by $-y$ yields $A^\top y \ge 0$, $y \ge 0$, and $b^\top y < 0$.

2.

We show that $\{x : Ax = b\}$ is nonempty if and only if $\{y : A^\top y = 0,\ b^\top y \ne 0\}$ is empty.

($\Rightarrow$) If $Ax = b$ and $A^\top y = 0$, then $b^\top y = x^\top A^\top y = 0$.

($\Leftarrow$) Let $L := \{Ax : x \in \mathbb{R}^n\}$, a subspace, hence a closed convex set. The system $Ax = b$ is infeasible if and only if $b \notin L$. By the strict separating hyperplane theorem there exist $y \in \mathbb{R}^m$ and $\alpha \in \mathbb{R}$ such that

$$y^\top z \le \alpha \quad \forall z \in L, \qquad y^\top b > \alpha.$$

Write $z = Ax$. Then $y^\top Ax \le \alpha$ for all $x \in \mathbb{R}^n$, i.e. $(A^\top y)^\top x \le \alpha$ for all x. Taking $\pm x$ forces $A^\top y = 0$, and $x = 0$ gives $\alpha \ge 0$. Thus $b^\top y > \alpha \ge 0$, so $b^\top y \ne 0$.